Lesson 6: Transformers

ChatGPT edition — isolated battery pulses only

Turns, voltage, current, and losses

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

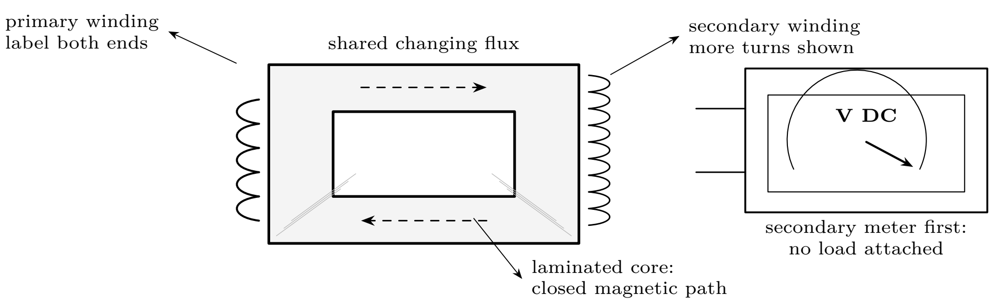
What you need

- Closed laminated iron core and two insulated low-voltage windings
- 2 AA cells in a holder; momentary pushbutton
- Two digital multimeters, insulated leads, tape labels
- Optional resistor load, 100 Ω or greater, rated 0.25 W

Predict first

When the primary current becomes steady, what remains on the secondary meter: a steady voltage, zero, or a changing voltage? What happens at release?

Read the apparatus before connecting it



DE-ENERGIZED CHECKS

With the battery holder disconnected: inspect insulation, identify each winding with continuity mode, record its resistance, and verify no continuity from either winding to the core. Stop if insulation is cut, a winding is hot, or a core-contact reading appears.

Check	N_p	R_p	N_s	R_s	Result / question
As labeled	_______	_______	_______	_______	Which winding has finer wire?
Core isolation	—		meter must show open circuit		○ pass ○ stop

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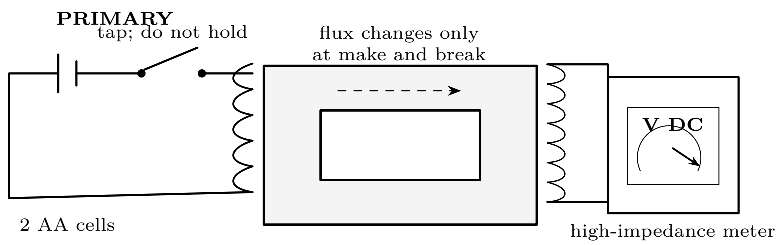
ChatGPT edition — staged verification

Turns, voltage, current, and losses

Stage 1: prove that change matters



1. Set the secondary meter to a safe DC-voltage range. Leave the secondary unloaded.
2. Tap the button briefly. Record the peak and sign at **make**; keep it pressed only long enough to see the reading return toward zero.
3. Release. Record the peak and sign at **break**. Repeat three times.
4. Remove the core, change nothing else, and repeat one make/break pair.

Trial	make V_s	held V_s	break V_s	core?	Evidence
1				yes	
2				yes	
3				yes	
4				no	

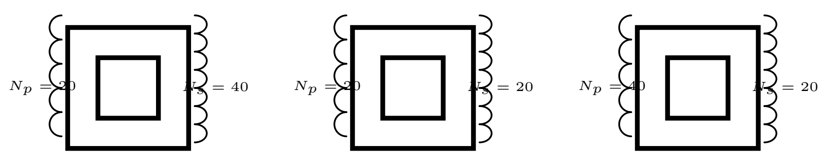
Questions before Stage 2

Why do make and break have opposite signs? Why does a steady battery current not give steady secondary voltage? What did the core change: turns, resistance, or magnetic coupling?

Stage 2: compare turns and voltage

Use the same tap duration and primary winding. Test secondaries one at a time.

$$\frac{V_s}{V_p} \approx \frac{N_s}{N_p}$$



N_p	N_s	N_s/N_p	V_p during tap	peak V_s	Does magnitude follow turns?
20	40	2.00			
20	20	1.00			
40	20	0.50			

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ChatGPT edition — ratio, load, and loss

Turns, voltage, current, and losses

Work the ideal ratios first

Example

For $N_p = 60$, $N_s = 180$, and a measured primary pulse of 1.8 V:

$$V_s = V_p \frac{N_s}{N_p} = 1.8\,\mathrm{V}\left(\frac{180}{60}\right) = 5.4\,\mathrm{V}.$$

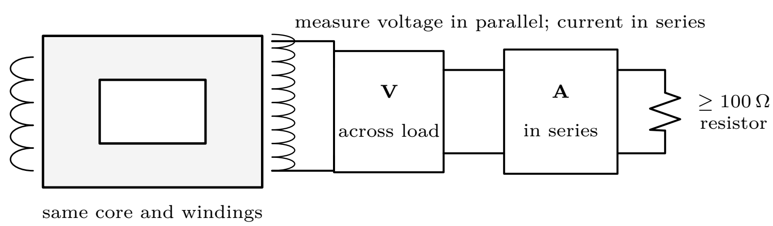
If an ideal secondary supplies 12 mA, power balance predicts

$$I_p \approx I_s \frac{N_s}{N_p} = 12\,\mathrm{mA}(3) = 36\,\mathrm{mA}.$$

Higher voltage comes with lower available current; a transformer does not create power.

N_p	N_s	V_p	predicted V_s	measured peak V_s
40	120	2.4 V	________	________
100	50	3.0 V	________	________
80	80	1.5 V	________	________

Stage 3: add one known load



1. Predict $I_s = V_s/R_L$. Have the adult check meter jacks and ranges.
2. Attach the resistor only after the open-circuit voltage test passes.
3. Use identical brief taps; record loaded voltage and current. Disconnect.
4. Calculate $P_s = V_s I_s$. Treat pulse readings as comparisons, not precision AC power measurements.

Condition	V_s	I_s	$V_s I_s$	winding warm?	stop?
Open circuit		—	—	no	
100 Ω load					

Where the ideal model loses

Loaded voltage falls because copper resistance causes I^2R heating, leakage flux misses the other winding, and the core dissipates energy through hysteresis and eddy currents. Shorted turns are faults, not experiments: they can draw high current and heat rapidly.

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Diagnose with power disconnected

Observation	Safe check	Likely explanation
No make/break pulse	continuity, meter range, core seated, switch action	open lead, wrong meter mode, poor coupling
Pulse only when probing	flex one lead gently; inspect terminals	intermittent connection
Much smaller than predicted	recount turns; compare with core removed	leakage, incomplete core, switching inconsistency
Loaded voltage collapses	remove load; verify resistor and meter placement	load too low, winding resistance, ammeter miswired
Heating, odor, damaged insulation	disconnect battery immediately	shorted winding, stalled current, wiring fault
Any winding-to-core continuity	stop; do not energize	insulation failure

FAULT-ISOLATION ORDER

Battery out. Inspect. Check winding continuity. Check isolation to the core. Verify meter leads and range. Restore only the open-circuit secondary test. Add the known load last. Never “see whether it clears” by holding the button.

Final proof: make four claims from evidence

1. **Changing flux:** show make and break pulses, opposite signs, and near-zero held reading.
2. **Magnetic coupling:** compare the same winding pair with and without the closed core.
3. **Turns ratio:** compare measured pulse ratios with N_s/N_p for at least two winding pairs.
4. **Non-ideal behavior:** show that a known load reduces secondary voltage and account for at least two loss mechanisms.

Claim	Best measurement	Because
Flux must change	____________________	____________________
Core couples coils	____________________	____________________
Turns set ratio	____________________	____________________
Losses are real	____________________	____________________

Exit questions

If $N_s/N_p = 4$, what happens ideally to voltage and available current? Why is a DC battery useful here only when switched? Which single measurement best distinguishes an open secondary from poor core coupling? Why must an ammeter never be placed directly across a winding?

Change one thing

Graph peak secondary voltage against turns while keeping primary, core, meter, and tap method fixed. Then choose one controlled comparison—core gap, load resistance, or winding orientation—and explain the result.

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.